

# VIJAY S

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## Professional Summary

B.Tech Information Technology graduate with hands-on experience in backend and full-stack development. Skilled in designing secure RESTful APIs, building scalable web applications, and developing real-time data dashboards. Strong foundation in data structures, database design, and backend development. Seeking an entry-level Software Engineer role to build efficient, production-ready systems.

## Education

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| <b>Sri Sairam Institute of Technology</b><br><i>B.Tech in Information Technology (CGPA: 8.08/10)</i>           | <b>2022 – 2026</b><br><i>Chennai</i> |
| <b>St. Marks Matriculation Hr. Sec. School</b><br><i>Higher Secondary – Mathematics &amp; Computer Science</i> | <b>2020 – 2022</b><br><i>Chennai</i> |

## Internship

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| <b>Chennai Unified Metropolitan Transport Authority (CUMTA)</b><br><i>Web Development Intern</i> | <b>Jun 2024 – Sep 2024</b><br><i>Chennai</i> |
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- Developed real-time dashboards to monitor vehicular movement and transport analytics, improving operational visibility for internal teams.
- Designed and integrated RESTful APIs using Python Flask to process and serve live transport data.
- Implemented secure role-based authentication and authorization, enhancing system access control.
- Built responsive front-end interfaces using React.js and Bootstrap, improving usability across devices.
- Optimized database queries to improve dashboard data retrieval performance.

## Projects

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|-------------------------------------------------------------|----------------------------------|
| <b>QR-Based Pavilion Tracking System   Next.js, MongoDB</b> | <b>National Event Deployment</b> |
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- Engineered a real-time tracking system managing 15,000+ attendees across 14 zones during a large-scale event.
- Implemented automated access control logic based on dynamic occupancy thresholds to prevent overcrowding.
- Developed analytics dashboards to monitor crowd flow, zone capacity, and live utilization metrics.
- Designed scalable backend APIs to handle concurrent QR validation requests.

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| <b>Intelligent Road Safety System   Python, YOLOv8, Twilio API</b> | <b>2024</b> |
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- Built an AI-based accident detection system using YOLOv8 for real-time CCTV video analysis.
- Integrated automated emergency alert notifications using Twilio API to reduce response time.
- Improved detection accuracy through model tuning and threshold optimization.

## Technical Skills

- Languages:** Java, Python, JavaScript
- Frontend:** React.js, Next.js, HTML, CSS, Bootstrap
- Backend:** Flask, RESTful APIs
- Libraries:** Pandas, NumPy
- Databases:** MySQL, PostgreSQL, MongoDB
- Tools:** Git, GitHub, Postman
- Core Concepts:** OOPS, DBMS, Data Structures & Algorithms